

Reading Arrays: Rows, Groups, and the Total

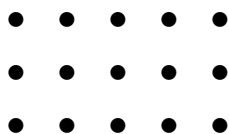
Grade 2–3 Multiplication

Three different questions — read carefully which one is asked:

- **How many rows?** → count the rows going down the side.
- **How many in each row?** → count across just one row.
- **How many in all (the total)?** → rows $\times$ in each row.

The total is almost always *bigger* than the number of rows. If your total equals the number of rows, you answered the wrong question.

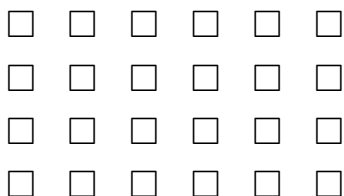
1. Look at this array of dots.



There are **3** rows with **5** dots in each row. How many dots are there **in all**?

Answer: _____

2. This array has **4** rows with **6** squares in each row.



Write the multiplication sentence for the picture, then find the **total number of squares**.

Answer: _____

3. A garden has **6** rows of carrots with **3** carrots in each row.

Circle the words that tell you the *total*, then find how many carrots are in the garden altogether.

Answer: _____

4. Find and fix the mistake. Jonah's array has **5** rows with **4** stickers in each row.

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★ ★ ★ ★
★ ★ ★ ★
★ ★ ★ ★
★ ★ ★ ★
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Jonah writes: "There are **5** stickers in all, because I count 5 rows."

What did Jonah count instead of the total? Write the correct total.

Answer: _____

5. Find and fix the mistake. Mia sets out **2** rows of shells with **8** shells in each row. She writes: $2 + 8 = 10$ shells in all.

Which operation should Mia have used, and what is the correct total number of shells?

Answer: _____

6. A baker puts **28** muffins into **4** equal rows.

How many muffins are in **each row**? (This is *not* 28 — that is the total for the whole tray.)

Answer: _____

7. Ella lines up **30** beads in rows with **5** beads in each row.

How many **rows** of beads does she make?

Answer: _____

8. Challenge. A checkerboard game uses **36** counters arranged in equal rows of **9** counters each. Sam says, “There must be 36 rows, because there are 36 counters.”

Explain in one sentence what Sam mixed up, then find the real number of rows.

Answer: _____